



SUMMARY OF PRODUCT CHARACTERISTICS
FORSTAVIR
Tenofovir disoproxil fumarate tablets 300 mg
Rx Only

1. NAME OF THE MEDICINAL PRODUCT
Tenofovir disoproxil fumarate tablets 300mg

2. QUALITATIVE AND QUANTITATIVE COMPOSITION
Each film-coated tenofovir disoproxil fumarate tablet contains 245 mg of tenofovir disoproxil (as fumarate), equivalent to 300 mg of tenofovir disoproxil fumarate, or 136 mg of tenofovir.
Each tablet contains 110.00 mg lactose monohydrate. For a full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM
Film-coated tablet.
Tenofovir disoproxil fumarate tablets are white to off-white, oval shaped, biconvex, film-coated tablets debossed with '1' on one side and '36' on the other side.

4.1 CLINICAL PARTICULARS
4.1 Therapeutic Indications
HIV-1 infection
Tenofovir disoproxil fumarate tablets is indicated in combination with other antiretroviral agents for the treatment of HIV-1 infection in adults and pediatric patients 12 years of age and older.
The following points should be considered while initiating therapy with Tenofovir disoproxil fumarate tablets for the treatment of HIV-1 infection:

- Tenofovir disoproxil fumarate tablets should not be used in combination with preparation containing emtricitabine/tenofovir disoproxil fumarate or efavirenz/emtricitabine/tenofovir disoproxil Fumarate.

Chronic Hepatitis B
Tenofovir disoproxil fumarate tablets is indicated for the treatment of chronic hepatitis B in adults.
The following points should be considered when initiating therapy with Tenofovir disoproxil fumarate tablets for the treatment of HBV infection:

- This indication is based primary on data from treatment of subjects who were nucleoside- treatment-naïve and a smaller number of subjects who had previously received lamivudine or adefovir dipivoxil. Subjects were adults with HBeAg-positive and HBeAg-negative chronic hepatitis with compensated liver disease.
- Tenofovir disoproxil fumarate tablet was evaluated in a limited number of subjects with chronic hepatitis B and decompensated liver disease.
- The numbers of subjects in clinical trials who had lamivudine- or adefovir-associated substitutions at baseline were too small to reach conclusions of efficacy.

4.2 Posology and method of administration
Recommended Dose in Adults
For the treatment of HIV-1 or chronic hepatitis B: The dose is one 300 mg Tenofovir disoproxil fumarate tablet once daily taken orally, without regard to food. In the treatment of chronic hepatitis B, the optimal duration of treatment is unknown.

Dose Adjustment for Renal Impairment in Adults
Significantly increased drug exposures occurred when Tenofovir disoproxil fumarate tablets was administered to subjects with moderate to severe renal impairment. Therefore, the dosing interval of Tenofovir disoproxil fumarate tablets should be adjusted in patients with baseline creatinine clearance <50 mL/min using the recommendations in Table 1. These dosing interval recommendations are based on modelling of single-dose pharmacokinetic data in non-HIV and non-HBV infected subjects with varying degrees of renal impairment, including end-stage renal disease requiring hemodialysis. The safety and effectiveness of these dosing interval adjustment recommendations have not been clinically evaluated in patients with moderate or severe renal impairment, therefore clinical response to treatment and renal function should be closely monitored in these patients.
No dose adjustment is necessary for patients with mild renal impairment (creatinine clearance 50–80 mL/min). Routine monitoring of calculated creatinine clearance and serum phosphorus should be performed in patients with mild renal impairment.

Table 1: Dosage Adjustment for Patients with Altered Creatinine Clearance

	Creatinine Clearance (mL/min) ^a			Hemodialysis Patients
	≥ 50	30-49	10-29	
Recommended 300 mg Dosing Interval	Every 24 hours	Every 48 hours	Every 72 to 96 hours	Every 7 days or after a total of approximately 12 hours of dialysis ^b

^a Calculated using ideal (lean) body weight
^b Generally once weekly assuming three hemodialysis sessions a week of approximately 4 hours duration.
Tenofovir disoproxil fumarate tablets should be administered following completion of dialysis.
The pharmacokinetics of tenofovir have not been evaluated in non-hemodialysis patients with creatinine clearance <10 mL/min; therefore, no dosing recommendation is available for these patients.
No data are available to make dose recommendations in pediatric patients 12 years of age and older with renal impairment.

4.3 Contraindications
Hypersensitivity to the active substance or to any of the excipients.
4.4 Special warnings and precautions for use
General:
Tenofovir disoproxil fumarate has not been studied in patients under the age of 12 or in patients over the age of 65 years. Elderly patients are more likely to have decreased renal function, therefore caution should be exercised when treating elderly patients with tenofovir disoproxil fumarate (see below).
HIV antibody testing should be offered to all HBV infected patients before initiating tenofovir disoproxil fumarate therapy (see below Co-infection with HIV–1 and hepatitis B).
Patients must be advised that tenofovir disoproxil fumarate has not been proven to prevent the risk of transmission of HIV or HBV to others through sexual contact or contamination with blood. Appropriate precautions must continue to be used.
Tenofovir disoproxil fumarate tablets contain lactose monohydrate. Consequently, patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency, or glucose–galactose malabsorption should not take this medicinal product.

Co-administration of other medicinal products:
- Tenofovir disoproxil fumarate tablets should not be administered with any other medicinal products containing tenofovir disoproxil fumarate.
- Tenofovir disoproxil fumarate tablets should also not be administered concurrently with adefovir dipivoxil.
- Co-administration of tenofovir disoproxil fumarate and didanosine is not recommended. Co-administration of tenofovir disoproxil fumarate and didanosine results in a 40-60% increase in systemic exposure to didanosine that may increase the risk of didanosine-related adverse events (see section 4.5). Rare cases of pancreatitis and lactic acidosis, sometimes fatal, have been reported. Co-administration of tenofovir disoproxil fumarate and didanosine at a dose of 400 mg daily has been associated with a significant decrease in CD4 cell count, possibly due to an intracellular interaction increasing phosphorylated (i.e. active) didanosine. A decreased dosage of 250 mg didanosine co-administered with tenofovir disoproxil fumarate therapy has been associated with reports of high rates of virological failure within several tested combinations for the treatment of HIV-1 infection.

Triple therapy with nucleosides/nucleotides: There have been reports of a high rate of virological failure and of emergence of resistance at early stage in HIV patients when tenofovir disoproxil fumarate was combined with lamivudine and abacavir as well as with lamivudine and didanosine as a once daily regimen.
Renal function: Tenofovir is principally eliminated via the kidney. Renal failure, renal impairment, elevated creatinine, hypophosphataemia and proximal tubulopathy (including Fanconi syndrome) have been reported with the use of tenofovir disoproxil fumarate in clinical practice (see section 4.8).
Renal safety with tenofovir has only been studied to a very limited degree in patients with impaired renal function (CrCl < 80 mL/min).

It is recommended that creatinine clearance is calculated in all patients prior to initiating therapy with tenofovir disoproxil fumarate and renal function (creatinine clearance and serum phosphate) is also monitored every four weeks during the first year, and then every three months. In patients at risk for renal impairment, including patients who have previously experienced renal events while receiving adefovir dipivoxil, consideration should be given to more frequent monitoring of renal function.
Patients with creatinine clearance < 50 mL/min, including haemodialysis patients: There are limited data on the safety and efficacy of tenofovir disoproxil fumarate in patients with impaired renal function. Therefore, tenofovir disoproxil fumarate should only be used if the potential benefits of treatment are considered to outweigh the potential risks. In patients with severe renal impairment (creatinine clearance < 30 mL/min) use of tenofovir is not recommended. If no alternative treatment is available, the dosing interval must be adjusted and renal function should be closely monitored (see sections 4.2 and 5.2).
If serum phosphate is < 1.5 mg/dl (0.48 mmol/l) or creatinine clearance is decreased to < 50 mL/min in any patient receiving tenofovir disoproxil fumarate, renal function should be re-evaluated within one week, including measurements of blood glucose, blood potassium and urine glucose concentrations (see section 4.8, proximal tubulopathy). Consideration should also be given to interrupting treatment with tenofovir disoproxil fumarate in patients with creatinine clearance decreased to < 50 mL/min or decreases in serum phosphate to < 1.0 mg/dl (0.32 mmol/l).
Use of tenofovir disoproxil fumarate should be avoided with concurrent or recent use of a nephrotoxic medicinal product (e.g. aminoglycosides, amphotericin B, foscamet, ganciclovir, pentamidine, vancomycin, didofvir or interleukin-2). If concomitant use of tenofovir disoproxil fumarate and nephrotoxic agents is unavoidable, renal function should be monitored weekly.

Tenofovir disoproxil fumarate has not been clinically evaluated in patients receiving medicinal products which are secreted by the same renal pathway, including the transport proteins human organic anion (hOAT) 1 and 3 or MRP 4 (e.g. didofvir, a known nephrotoxic medicinal product). These renal transport proteins may be responsible for tubular secretion and in part, renal elimination of tenofovir and didofvir. Consequently, the pharmacokinetics of these medicinal products which are secreted by the same renal pathway including transport proteins hOAT 1 and 3 or MRP 4 might be modified if they are co-administered. Unless clearly necessary, concomitant use of these medicinal products which are secreted by the same renal pathway is not recommended, but if such use is unavoidable, renal function should be monitored weekly (see section 4.5).
Bone effects: In HIV infected patients, in a 144-week controlled clinical study that compared tenofovir disoproxil fumarate with stavudine in combination with lamivudine and efavirenz in antiretroviral-naïve patients, small decreases in bone mineral density of the hip and spine were observed in both treatment groups. Decreases in bone mineral density of spine and changes in bone biomarkers from baseline were significantly greater in the tenofovir disoproxil fumarate treatment group at 144 weeks. Decreases in bone mineral density of hip were significantly greater in this group until 96 weeks. However, there was no increased risk of fractures or evidence for clinically relevant bone abnormalities over 144 weeks.

Bone abnormalities (infrequently contributing to fractures) may be associated with proximal renal tubulopathy (see section 4.8). If bone abnormalities are suspected then appropriate consultation should be obtained.
Liver disease: The safety of tenofovir in patients with decompensated liver disease is being studied. At present the safety in this patient population has not been thoroughly evaluated.
No safety and efficacy data are available in liver transplant patients.

Exacerbations of hepatitis:
Flares on treatment: Spontaneous exacerbations in chronic hepatitis B are relatively common and are characterised by transient increases in serum ALT. After initiating antiviral therapy, serum ALT may increase in some patients as serum HBV DNA levels decline (see section 4.8).
Among tenofovir-treated patients on-treatment exacerbations typically occurred after 4–8 weeks of therapy. In patients with compensated liver disease, these increases in serum ALT are generally not accompanied by an increase in serum bilirubin concentrations or hepatic decompensation. Patients with cirrhosis may be at a higher risk for hepatic decompensation following hepatitis exacerbation, and therefore should be monitored closely during therapy.
Flares after treatment discontinuation: Acute exacerbation of hepatitis has also been reported in patients who have discontinued hepatitis B therapy. Post-treatment exacerbations are usually associated with rising HBV DNA, and the majority appears to be self-limited. However, severe exacerbations, including fatalities, have been reported. Hepatic function should be monitored at repeated intervals with both clinical and laboratory follow-up for at least 6 months after discontinuation of hepatitis B therapy. If appropriate, resumption of hepatitis B therapy may be warranted. In patients with advanced liver disease or cirrhosis, treatment discontinuation is not recommended since post-treatment exacerbation of hepatitis may lead to hepatic decompensation.
Liver flares are especially serious, and sometimes fatal in patients with decompensated liver disease.
Co-infection with hepatitis C or D: There are no data on the efficacy of tenofovir in patients co-infected with hepatitis C or D virus.
Co-infection with HIV–1 and hepatitis B: Due to the risk of development of HIV resistance, tenofovir disoproxil fumarate should only be used as part of an appropriate antiretroviral combination regimen in HIV/HBV co-infected patients. Patients with pre-existing liver dysfunction including chronic active hepatitis have an increased frequency of liver function abnormalities during combination antiretroviral therapy and should be monitored according to standard practice. If there is evidence of worsening liver disease in such patients, interruption or discontinuation of treatment must be considered. However, it should be noted that increases of ALT can be part of HBV clearance during therapy with tenofovir, see above Exacerbations of hepatitis.

Lactic acidosis: Lactic acidosis, usually associated with hepatic steatosis, has been reported with the use of nucleoside analogues. The preclinical and clinical data suggest that the risk of occurrence of lactic acidosis, a class effect of nucleoside analogues, is low for tenofovir disoproxil fumarate. However, as tenofovir is structurally related to nucleoside analogues, this risk cannot be excluded. Early symptoms (symptomatic hyperlactatemia) include benign digestive symptoms (nausea, vomiting and abdominal pain), non-specific malaise, loss of appetite, weight loss, respiratory symptoms (rapid and/or deep breathing) or neurological symptoms (including motor weakness). Lactic acidosis has a high mortality and may be associated with pancreatitis, liver failure or renal failure. Lactic acidosis generally occurred after a few or several months of treatment.
Treatment with nucleoside analogues should be discontinued in the setting of symptomatic hyperlactatemia and metabolic/lactic acidosis, progressive hepatomegaly, or rapidly elevating aminotransferase levels.
Caution should be exercised when administering nucleoside analogues to any patient (particularly obese women) with hepatomegaly, hepatitis or other known risk factors for liver disease and hepatic steatosis (including certain medicinal products and alcohol). Patients co-infected with hepatitis C and treated with alpha interferon and ribavirin may constitute a special risk.
Patients at increased risk should be followed closely.

Lipodystrophy (lipatrophy/lipomatosis): In HIV infected patients, combination antiretroviral therapy has been associated with the redistribution of body fat (lipodystrophy). The long-term consequences of these events are currently unknown. Knowledge about the mechanism is incomplete. A connection



between visceral lipomatosis and protease inhibitors and lipatrophy and nucleoside reverse transcriptase inhibitors has been hypothesised. A higher risk of lipodystrophy has been associated with individual factors such as older age, and with drug related factors such as longer duration of antiretroviral treatment and associated metabolic disturbances. Clinical examination should include evaluation for physical signs of fat redistribution. Consideration should be given to the measurement of fasting serum lipids and blood glucose. Lipid disorders should be managed as clinically appropriate (see section 4.8).
Tenofovir is structurally related to nucleoside analogues hence the risk of lipodystrophy cannot be excluded. However, 144-week clinical data from antiretroviral-naïve HIV infected patients indicate that the risk of lipodystrophy was lower with tenofovir disoproxil fumarate than with stavudine when administered with lamivudine and efavirenz.

Mitochondrial dysfunction: Nucleoside and nucleotide analogues have been demonstrated *in vitro* and *in vivo* to cause a variable degree of mitochondrial damage. There have been reports of mitochondrial dysfunction in HIV negative infants exposed *in utero* and/or postnatally to nucleoside analogues. The main adverse events reported are haematological disorders (anaemia, neutropenia), metabolic disorders (hyperlactataemia, hyperlipasaemia). These events are often transitory. Some late-onset neurological disorders have been reported (hyperreflexia, convulsion, abnormal behaviour). Whether the neurological disorders are transient or permanent is currently unknown. Any child exposed *in utero* to nucleoside and nucleotide analogues, even HIV negative children, should have clinical and laboratory follow-up and should be fully investigated for possible mitochondrial dysfunction in case of relevant signs or symptoms. These findings do not affect current national recommendations to use antiretroviral therapy in pregnant women to prevent vertical transmission of HIV.

Immune Reactivation Syndrome: In HIV infected patients with severe immune deficiency at the time of institution of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of CART. Relevant examples are cytomegalovirus reinitis, generalised and/or focal mycobacterium infections, and *Pneumocystis jiroveci* pneumonia. Any inflammatory symptoms should be evaluated and treatment instituted when necessary.

Osteonecrosis: Although the etiology is considered to be multifactorial (including corticosteroid use, alcohol consumption, severe immunosuppression, higher body mass index), cases of osteonecrosis have been reported particularly in patients with advanced HIV-disease and/or long-term exposure to combination antiretroviral therapy (CART). Patients should be advised to seek medical advice if they experience joint aches and pain, joint stiffness, or difficulty in movement.

4.5 Interaction with other medicinal products and other forms of interaction
Interaction studies have only been performed in adults. Based on the results of *in vitro* experiments and the known elimination pathway of tenofovir, the potential for CYP450 mediated interactions involving tenofovir with other medicinal products is low.
Concomitant use not recommended:
Tenofovir disoproxil fumarate tablets should not be administered with any other medicinal products containing tenofovir disoproxil fumarate.
Tenofovir disoproxil fumarate tablets should also not be administered concurrently with adefovir dipivoxil.
Didanosine: Co-administration of tenofovir disoproxil fumarate and didanosine is not recommended (see section 4.4 and Table 1).
Renally eliminated medicinal products: Since tenofovir is primarily eliminated by the kidneys, co-administration of tenofovir disoproxil fumarate with medicinal products that reduce renal function or compete for active tubular secretion via transport proteins hOAT 1, hOAT 3 or MRP 4 (e.g. didofvir) may increase serum concentrations of tenofovir and/or the co-administered medicinal products.
Use of tenofovir disoproxil fumarate should be avoided with concurrent or recent use of a nephrotoxic medicinal product. Some examples include, but are not limited to, aminoglycosides, amphotericin B, foscamet, ganciclovir, pentamidine, vancomycin, didofvir or interleukin-2 (see section 4.4).
Given that tacrolimus can affect renal function, close monitoring is recommended when it is co-administered with tenofovir disoproxil fumarate.

Other interactions:
Interactions between tenofovir disoproxil fumarate and protease inhibitors and antiretroviral agents other than protease inhibitors are listed in Table 1 below (increase is indicated as “↑”, decrease as “↓”, no change as “↔”, twice daily as “b.i.d.”, and once daily as “q.d.”).
Table 1: Interactions between tenofovir disoproxil fumarate and other medicinal products

Medicinal product by therapeutic areas (dose in mg)	Effects on drug levels Mean percent change in AUC, C _{max} , C _{min}	Recommendation concerning co-administration with tenofovir disoproxil fumarate 300 mg
ANTI-INFECTIVES		
Antiretrovirals		
Protease inhibitors		
Atazanavir/Ritonavir (300 q.d./100 q.d./300 q.d.)	Atazanavir: AUC: ↓ 25% C _{max} : ↓ 28% C _{min} : ↓ 26% Tenofovir: AUC: ↑ 37% C _{max} : ↑ 34% C _{min} : ↑ 29%	No dose adjustment is recommended. The increased exposure of tenofovir could potentiate tenofovir associated adverse events, including renal disorders. Renal function should be closely monitored (see section 4.4).
Lopinavir/Ritonavir (400 b.i.d./100 b.i.d./300 q.d.)	Lopinavir/ritonavir: No significant effect on lopinavir/ritonavir PK parameters. Tenofovir: AUC: ↑ 32% C _{max} : ↔ C _{min} : ↑ 51%	No dose adjustment is recommended. The increased exposure of tenofovir could potentiate tenofovir associated adverse events, including renal disorders. Renal function should be closely monitored (see section 4.4).
Darunavir/Ritonavir (300/100 b.i.d./300 q.d.)	Darunavir: No significant effect on darunavir/ritonavir PK parameters. Tenofovir: AUC: ↑ 22% C _{min} : ↑ 37%	No dose adjustment is recommended. The increased exposure of tenofovir could potentiate tenofovir associated adverse events, including renal disorders. Renal function should be closely monitored (see section 4.4).
NRTIs		
Didanosine	Co-administration of tenofovir disoproxil fumarate and didanosine results in a 40–60% increase in systemic exposure to didanosine that may increase the risk for didanosine-related adverse events. Rare cases of pancreatitis and lactic acidosis, sometimes fatal, have been reported. Co-administration of tenofovir disoproxil fumarate and didanosine at a dose of 400 mg daily has been associated with a significant decrease in CD4 cell count, possibly due to an intracellular interaction increasing phosphorylated (i.e. active) didanosine. A decreased dosage of 250 mg didanosine co-administered with tenofovir disoproxil fumarate therapy has been associated with reports of high rates of virological failure within several tested combinations for the treatment of HIV–1 infection.	Co-administration of tenofovir disoproxil fumarate and didanosine is not recommended (see section 4.4).
Adefovir dipivoxil	AUC: ↔ C _{max} : ↔	Tenofovir disoproxil fumarate should not be administered concurrently with adefovir dipivoxil (see section 4.4).
Entecavir	AUC: ↔ C _{max} : ↔	No clinically significant pharmacokinetic interactions when tenofovir disoproxil fumarate was co-administered with entecavir.

Studies conducted with other medicinal products: There were no clinically significant pharmacokinetic interactions when tenofovir disoproxil fumarate was co-administered with emtricitabine, lamivudine, indinavir, efavirenz, nelfinavir, saquinavir (ritonavir boosted), methadone, ribavirin, rifampicin, tacrolimus, or the hormonal contraceptive norgestimate/ethinyl oestradiol.
Tenofovir disoproxil fumarate must be taken with food, as food enhances the bioavailability of tenofovir (see section 5.2).

4.6 Fertility, Pregnancy and lactation
Pregnancy
For tenofovir disoproxil fumarate limited clinical data on exposed pregnancies are available.
Animal studies do not indicate direct or indirect harmful effects of tenofovir disoproxil fumarate with respect to pregnancy, foetal development, parturition or postnatal development (see section 5.3).
Tenofovir disoproxil fumarate should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.
Given that the potential risks to developing human foetuses are unknown, the use of tenofovir disoproxil fumarate in women of childbearing potential must be accompanied by the use of effective contraception.

Lactation
In animal studies it has been shown that tenofovir is excreted into milk. It is not known whether tenofovir is excreted in human milk. Therefore, it is recommended that mothers being treated with tenofovir disoproxil fumarate should not breast-feed their infants.
As a general rule, it is recommended that HIV and HBV infected women do not breast-feed their infants in order to avoid transmission of HIV and HBV to the infant.
Fertility
There are limited clinical data with respect to the effect of tenofovir disoproxil on fertility. Animal studies do not indicate harmful effects of tenofovir disoproxil on fertility.

4.7 Effects on ability to drive and use machines
No studies on the effects on the ability to drive and use machines have been performed. However, patients should be informed that dizziness has been reported during treatment with tenofovir disoproxil fumarate.
4.8 Undesirable effects
Approximately one third of patients can be expected to experience adverse reactions following treatment with tenofovir disoproxil fumarate in combination with other antiretroviral agents. These reactions are usually mild to moderate gastrointestinal events.
The adverse reactions with suspected (at least possible) relationship to treatment are listed below by body system organ class and absolute frequency. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness. Frequencies are defined as very common (≥ 1/10) or common (≥ 1/100, < 1/10).

Metabolism and nutrition disorders:
Very common: hypophosphataemia
Uncommon: hypokalaemia
Rare: lactic acidosis
Nervous system disorders:
Very common: dizziness
Common: headache
Gastrointestinal disorders:
Very common: diarrhoea, vomiting, nausea
Common: abdominal pain, abdominal distension, flatulence
Uncommon: pancreatitis
Hepatobiliary disorders:
Common: increased transaminases
Rare: hepatic steatosis, hepatitis
Skin and subcutaneous tissue disorders:
Very common: rash
Rare: Angioedema
Musculoskeletal and connective tissue disorders:
Common: bone mineral density decreased
Uncommon: rhabdomyolysis, muscular weakness
Rare: osteomalacia (manifested as bone pain and infrequently contributing to fractures), myopathy
Renal and urinary disorders:
Uncommon: increase creatinine, proximal renal tubulopathy (including Fanconi syndrome)
Rare: acute renal failure, renal failure, acute tubular necrosis, nephritis (including acute interstitial nephritis), nephrogenic diabetes insipidus.
General disorders and administration site conditions:
Very common: asthenia
Common: fatigue
Approximately 1% of tenofovir disoproxil fumarate-treated patients discontinued treatment due to the gastrointestinal events.
Combination antiretroviral therapy has been associated with metabolic abnormalities such as hypertriglyceridaemia, hypercholesterolaemia, insulin

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	Product Name	Component	Item Code	Date & Time
	FORSTAVIR Tablets	Leaflet	P1541966	10.03.2026 & 02.35 pm
	Country	Version No.	Reason Of Issue	Reviewed / Approved by
	Malaysia - U3	00	New	
Team Leader	Lova raju	Dimensions (mm)	Colours: 01	
Initiator	Subramanyam	350 x 540 mm		
Artist	Ashoki	Pharma Code: 41966		
Additional Information:				

resistance, hyperglycaemia and hyperlactataemia (see section 4.4).

Combination antiretroviral therapy has been associated with redistribution of body fat (lipodystrophy) in HIV patients including the loss of peripheral and facial subcutaneous fat, increased intra-abdominal and visceral fat, breast hypertrophy and dorsocervical fat accumulation (buffalo hump).

In a 144-week controlled clinical study in antiretroviral-naïve patients that compared tenofovir disoproxil fumarate with stavudine in combination with lamivudine and efavirenz, patients who received tenofovir disoproxil had a significantly lower incidence of lipodystrophy compared with patients who received stavudine. The tenofovir disoproxil fumarate arm also had significantly smaller mean increases in fasting triglycerides and total cholesterol than the comparator arm.

In HIV infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise (see section 4.4).

Cases of osteonecrosis have been reported, particularly in patients with generally acknowledged risk factors, advanced HIV disease or long-term exposure to combination antiretroviral therapy (CART). The frequency of this is unknown (see section 4.4).

Hepatitis B: Assessment of adverse reactions from clinical study data is based on experience in two double-blind comparative-controlled studies in which 641 patients with chronic hepatitis B and compensated liver disease received treatment with tenofovir disoproxil 245 mg (as fumarate) daily (n = 426) or adefovir dipivoxil 10 mg daily (n = 215) for 48 weeks.

Exacerbations during treatment:

In studies with nucleoside-naïve patients, on-treatment ALT elevations > 10 times ULN (upper limit of normal) and > 2 times baseline occurred in 2.6% of tenofovir disoproxil fumarate-treated patients versus 1.9% of adefovir dipivoxil-treated patients.

ALT elevations had a median time to onset of 8 weeks, resolved with continued treatment, and, in a majority of cases, were associated with a ≥ 2 log copies/ml reduction in viral load that preceded or coincided with 10 the ALT elevation. Periodic monitoring of hepatic function is recommended during treatment.

The following adverse reactions, listed under the body system headings above, may occur as a consequence of proximal renal tubulopathy: rhabdomyolysis, osteomalacia (manifested as bone pain and infrequently contributing to fractures), hypokalaemia, muscular weakness, myopathy and hypophosphataemia. These events are not considered to be causally associated with tenofovir disoproxil fumarate therapy in the absence of proximal renal tubulopathy.

4.9 Overdose

If overdose occurs the patient must be monitored for evidence of toxicity (see sections 4.8 and 5.3), and standard supportive treatment applied as necessary.

Tenofovir can be removed by haemodialysis; the median haemodialysis clearance of tenofovir is 134 ml/min. The elimination of tenofovir by peritoneal dialysis has not been studied.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Nucleoside and nucleotide reverse transcriptase inhibitors, ATC code: J05AF07

Mechanism of action: Tenofovir disoproxil fumarate is the fumarate salt of the prodrug tenofovir disoproxil. Tenofovir disoproxil is absorbed and converted to the active substance tenofovir, which is a nucleoside monophosphate (nucleotide) analogue. Tenofovir is then converted to the active diphosphate, tenofovir diphosphate, an obligate chain terminator, by constitutively expressed cellular enzymes. Tenofovir diphosphate has an intracellular half-life of 10 hours in activated and 50 hours in resting peripheral blood mononuclear cells (PBMCs). Tenofovir diphosphate inhibits HIV-1 reverse transcriptase and the HBV polymerase by direct binding competition with the natural deoxyribonucleotide substrate and, after incorporation into DNA, by DNA chain termination. Tenofovir diphosphate is a weak inhibitor of cellular polymerases α, β, and γ. At concentrations of up to 300 μmol/l, tenofovir has also shown no effect on the synthesis of mitochondrial DNA or the production of lactic acid *in vitro* assays.

Data pertaining to HBV:

HIV antiviral activity *in vitro*: The concentration of tenofovir required for 50% inhibition (EC₅₀) of the wild-type laboratory strain HIV-1_{lab} is 1–6 μmol/l in lymphoid cell lines and 1.1 μmol/l against primary HIV-1 subtype B isolates in PBMCs. Tenofovir is also active against HIV-1 subtypes A, C, D, E, F, G, and O and against HIV_{lab} in primary monocyte/macrophage cells. Tenofovir shows activity *in vitro* against HIV-2, with an EC₅₀ of 4.9 μmol/l in MT-4 cells.

Resistance: Strains of HIV-1 with reduced susceptibility to tenofovir and a K65R mutation in reverse transcriptase have been selected *in vitro* and in some patients (see Clinical results).

Tenofovir disoproxil fumarate should be avoided in antiretroviral experienced patients with strains harbouring the K65R mutation (see section 4.4).

Clinical studies in treatment-experienced patients have assessed the anti-HIV activity of tenofovir disoproxil 245 mg (as fumarate) against strains of HIV-1 with resistance to nucleoside inhibitors. The results indicate that patients whose HIV expressed 3 or more thymidine-analogue associated mutations (TAMs) that included either the M41L or L210W reverse transcriptase mutation showed reduced response to tenofovir disoproxil 245 mg (as fumarate) therapy.

Clinical results: The effects of tenofovir disoproxil fumarate in treatment-experienced and treatment-naïve HIV-1 infected adults have been demonstrated in trials of 48 weeks duration in treatment-experienced HIV-1 infected adults.

In study GS-99-907, 550 treatment-experienced patients were treated with placebo or tenofovir disoproxil 245 mg (as fumarate) for 24 weeks. The mean baseline CD4 cell count was 427 cells/mm³; the mean baseline plasma HIV-1 RNA was 3.4 log₁₀ copies/ml; 78% of patients had a viral 10 fold of < 5,000 copies/ml and the mean duration of prior HIV treatment was 5.4 years. Baseline genotypic analysis of HIV isolates from 253 patients revealed that 94% of patients had HIV-1 resistance mutations associated with nucleoside reverse transcriptase inhibitors, 58% had mutations associated with protease inhibitors and 48% had mutations associated with non-nucleoside reverse transcriptase inhibitors.

At week 24 the time-weighted average change from baseline in log₁₀ plasma HIV-1 RNA levels (DAVG₂₄) was –0.03 log₁₀ copies/ml and –0.61 log₁₀ copies/ml for the placebo and tenofovir disoproxil 245 mg (as fumarate) recipients (p < 0.0001). A statistically significant difference in favour of tenofovir disoproxil 245 mg (as fumarate) was seen in the time-weighted average change from baseline at week 24 (DAVG₂₄) for CD4 count (+13 cells/mm³ for tenofovir disoproxil 245 mg (as fumarate) versus –11 cells/mm³ for placebo, p-value = 0.0008). The antiviral response to tenofovir disoproxil fumarate was durable through 48 weeks (DAVG₄₈ was –0.57 log₁₀ copies/ml; proportion of patients with HIV-1 RNA below 400 or 50 copies/ml was 41% and 18% respectively). Eight (2%) tenofovir disoproxil 245 mg (as fumarate) treated patients developed the K65R mutation within the first 48 weeks.

The 144-week, double-blind, active controlled phase of study GS-99-903 evaluated the efficacy and safety of tenofovir disoproxil 245 mg (as fumarate) versus stavudine when used in combination with lamivudine and efavirenz in HIV-1 infected patients naïve to antiretroviral therapy. The mean baseline CD4 cell count was 279 cells/mm³, the mean baseline plasma HIV-1 RNA was 4.91 log₁₀ copies/ml, 19% of patients had symptomatic HIV-1 infection and 18% had AIDS. Patients were stratified by baseline HIV-1 RNA and CD4 count. Forty-three percent of patients had baseline viral loads > 100,000 copies/ml and 39% had CD4 cell counts < 200 cells/ml.

By intent to treat analysis (Missing data and switch in antiretroviral therapy (ART) considered as failure), the proportion of patients with HIV-1 RNA below 400 copies/ml and 50 copies/ml at 48 weeks of treatment was 80% and 76% respectively in the tenofovir disoproxil 245 mg (as fumarate) arm, compared to 84% and 80% in the stavudine arm. At 144 weeks, the proportion of patients with HIV-1 RNA below 400 copies/ml and 50 copies/ml was 71% and 58% respectively in the tenofovir disoproxil 245 mg (as fumarate) arm, compared to 64% and 63% in the stavudine arm.

The average change from baseline for HIV-1 RNA and CD4 count at 48 weeks of treatment was similar in both treatment groups (–3.09 and –3.09 log₁₀ copies/ml; +169 and 167 cells/mm³ in the tenofovir disoproxil 245 mg (as fumarate) and stavudine groups, respectively). At 144 weeks of treatment, the average change from baseline remained similar in both treatment groups (–3.07 and –3.03 log₁₀ copies/ml; +263 and +283 cells/mm³ in the tenofovir disoproxil 245 mg (as fumarate) and stavudine groups, respectively). A consistent response to treatment with tenofovir disoproxil 245 mg (as fumarate) was seen regardless of baseline HIV-1 RNA and CD4 count.

The K65R mutation occurred in a slightly higher percentage of patients in the tenofovir disoproxil fumarate group than the active control group (2.7% versus 0.7%). Efavirenz or lamivudine resistance either preceded or was coincident with the development of K65R in all cases. Eight patients had HIV that expressed K65R in the tenofovir disoproxil 245 mg (as fumarate) arm, 7 of these occurred during the first 48 weeks of treatment and the last one at week 96. No further K65R development was observed up to week 144. From both the genotypic and phenotypic analyses there was no evidence for other pathways of resistance to tenofovir.

Data pertaining to HBV:

HBV antiviral activity *in vitro*: The *in vitro* antiviral activity of tenofovir against HBV was assessed in the HepG2 2.2.15 cell line. The EC₅₀ values for tenofovir were in the range of 0.14 to 1.5 μmol/l, with CC₅₀ (50% cytotoxicity concentration) values > 100 μmol/l.

Resistance: No HBV mutations associated with tenofovir disoproxil fumarate resistance have been identified (see *Clinical results*). In cell based assays, HBV strains expressing the rT173L, rT180M, and rM204I/V mutations associated with resistance to lamivudine and telbivudine showed a susceptibility to tenofovir ranging from 0.7- to 3.4-fold that of wild-type virus. HBV strains expressing the rT180M, rT184G, rS202G/I, rM204V and rM250V mutations associated with resistance to entecavir showed a susceptibility to tenofovir ranging from 0.6- to 6.9-fold that of wild-type virus. HBV strains expressing the adefovir-associated resistance mutations rT181V and rN236T showed a susceptibility to tenofovir ranging from 2.9- to 10-fold that of wild-type virus. Viruses containing the rT181T mutation remained susceptible to tenofovir with EC₅₀ values 1.5-fold that of wild-type virus.

Clinical results: The demonstration of benefit of tenofovir disoproxil fumarate is based on histological, virological, biochemical and serological responses mainly in treatment-naïve adults with HBeAg positive and HBeAg negative chronic hepatitis B with compensated liver disease.

Experience in patients with compensated liver disease: Results through 48 weeks from two randomised, phase 3 double-blind studies comparing tenofovir disoproxil fumarate to adefovir dipivoxil in patients with compensated liver disease are presented in Table 2 below. Study GS-US-174-0103 was conducted in 266 (randomised and treated) HBeAg positive patients while study GS-US-174-0102 was conducted in 375 (randomised and treated) patients negative for HBeAg and positive for HBeAb.

In both of these studies tenofovir disoproxil fumarate was significantly superior to adefovir dipivoxil for the primary efficacy endpoint of complete response (defined as HBV DNA levels < 400 copies/ml and Knodell necroinflammatory score improvement of at least 2 points without worsening in Knodell fibrosis). Treatment with tenofovir disoproxil 245 mg (as fumarate) was also associated with significantly greater proportions of patients with HBV DNA < 400 copies/ml, when compared to adefovir dipivoxil 10 mg treatment. Both treatments produced similar results with regard to histological response (defined as Knodell necroinflammatory score improvement of at least 2 points without worsening in Knodell fibrosis) at week 48 (see Table 2 below).

In study GS-US-174-0103 a significantly greater proportion of patients in the tenofovir disoproxil fumarate group than in the adefovir dipivoxil group had normalised ALT and achieved HBsAg loss at week 48 (see Table 2 below).

Table 2: Efficacy parameters in compensated HBeAg positive and HBeAg negative patients at week 48

	Study 174-0102 (HBeAg negative)		Study 174-0103 (HBeAg positive)	
Parameter	Tenofovir disoproxil 245 mg (as fumarate) n = 250	Adefovir dipivoxil 10 mg n = 125	Tenofovir disoproxil 245 mg (as fumarate) n = 176	Adefovir dipivoxil 10 mg n = 90
Complete response (%)^a	71*	49	67*	12
Histology				
Histological response (%) ^b	72	69	74	68
Median HBV DNA reduction from baseline^c (log ₁₀ copies/ml)	-4.7*	-4.0	-6.4*	-3.7
HBV DNA (%)				
< 400 copies/ml (< 69 IU/ml)	93*	63	76*	13
ALT (%)				
Normalised ALT ^d	76	77	68*	54
Serology (%)				
HBeAg loss/seroconversion	N/A	N/A	22/21	18/18
HBsAg loss/seroconversion	0/0	0/0	3*/1	0/0

* p-value versus adefovir dipivoxil < 0.05.

^a Complete response defined as HBV DNA levels < 400 copies/ml and Knodell necroinflammatory score improvement of at least 2 points without worsening in Knodell fibrosis.

^b Knodell necroinflammatory score improvement of at least 2 points without worsening in Knodell fibrosis.

^c Median change from baseline HBV DNA merely reflects the difference between baseline HBV DNA and the limit of detection (LOD) of the assay.

^d The population used for analysis of ALT normalisation included only patients with ALT above ULN at baseline.

Tenofovir disoproxil fumarate was associated with significantly greater proportions of patients with undetectable HBV DNA (< 169 copies/ml [$\leq$ 29 IU/ml]; the limit of quantification of the Roche Cobas Taqman HBV assay), when compared to adefovir dipivoxil (study GS-US-174-0102; 91%, 56% and study GS-US-174-0103; 69%, 9%), respectively.

Response to treatment with tenofovir disoproxil fumarate was comparable in nucleoside-experienced (n = 51) and nucleoside-naïve (n = 375) patients and in patients with normal ALT (n = 21) and abnormal ALT (n = 405) at baseline when studies GS-US-174-0102 and GS-US-174-0103 were combined. Forty-nine of the 51 nucleoside-experienced patients were previously treated with lamivudine. Seventy-three percent of nucleoside-experienced and 69% of nucleoside-naïve patients achieved complete response to treatment; 90% of nucleoside-naïve and 88% of nucleoside-experienced patients achieved HBV DNA suppression < 400 copies/ml. All patients with normal ALT at baseline and 88% of patients with abnormal ALT at baseline achieved HBV DNA suppression < 400 copies/ml.

In a randomised, 48-week double-blind, controlled study of tenofovir disoproxil 245 mg (as fumarate) in patients co-infected with HIV-1 and chronic hepatitis B with prior lamivudine experience (study ACTG 5127), the mean serum HBV DNA levels at baseline in patients randomised to the tenofovir arm were 9.45 log₁₀ copies/ml (n = 27). Treatment with tenofovir disoproxil 245 mg (as fumarate) was associated with a mean change in serum HBV DNA from baseline, in the patients for whom there was 48-week data, of –5.74 log₁₀ copies/ml (n = 18). In addition, 61% of patients had normal ALT at week 48.

Experience in patients with persistent viral replication: The efficacy and safety of tenofovir disoproxil 245 mg (as fumarate) or tenofovir disoproxil 245 mg (as fumarate) plus 200 mg emtricitabine is being evaluated in a randomised, double-blind study (study GS-US-174-0106), in HBeAg positive and HBeAg negative patients who had persistent viraemia (HBV DNA ≥ 1,000 copies/ml) while receiving adefovir dipivoxil 10 mg for more than 24 weeks. Overall at week 24, treatment with a tenofovir-containing regimen resulted in 77% (46/60) of patients with HBV DNA < 400 copies/ml (< 69 IU/ml) and 62% (37/60) of patients with undetectable HBV DNA (below 169 copies/ml [$\leq$ 29 IU/ml]; the limit of quantification of the Roche Cobas TaqMan HBV assay). In addition, at week 24, the percentage of patients who had normal ALT was 68% (42/62).

Clinical resistance: Four hundred and twenty-six HBeAg negative (GS-US-174-0102, n = 250) and HBeAg positive (GS-US-174-0103, n = 176) patients were evaluated for genotypic changes in HBV polymerase. Genotypic evaluations were performed on all patients with HBV DNA> 400 copies/ml at week

48. No mutations associated with tenofovir disoproxil fumarate resistance have been identified.

5.2 Pharmacokinetic properties

Tenofovir disoproxil fumarate is a water-soluble ester prodrug which is rapidly converted *in vivo* to tenofovir and formaldehyde.

Tenofovir is converted intracellularly to tenofovir monophosphate and to the active component, tenofovir diphosphate.

Absorption

Following oral administration of tenofovir disoproxil fumarate to HIV infected patients, tenofovir disoproxil fumarate is rapidly absorbed and converted to tenofovir. Administration of multiple doses of tenofovir disoproxil fumarate with a meal to HIV infected patients resulted in mean (%CV) tenofovir C_{max}, AUC_{0-∞}, and C_{min} values of 326 (36.6%) ng/ml, 3,324 (41.2%) ng·h/ml and 64.4 (39.4%) ng/ml, respectively. Maximum tenofovir concentrations are observed in serum within one hour of dosing in the fasted state and within two hours when taken with food. The oral bioavailability of tenofovir from tenofovir disoproxil fumarate in fasted patients was approximately 25%. Administration of tenofovir disoproxil fumarate with a high fat meal enhanced the oral bioavailability, with an increase in tenofovir AUC by approximately 40% and C_{max} by approximately 14%. Following the first dose of tenofovir disoproxil fumarate in fed patients, the median C_{min} in serum ranged from 213 to 375 ng/ml. However, administration of tenofovir disoproxil fumarate with a light meal did not have a significant effect on the pharmacokinetics of tenofovir.

Distribution

Following intravenous administration, the steady-state volume of distribution of tenofovir was estimated to be approximately 800 ml/kg. After oral administration of tenofovir disoproxil fumarate, tenofovir is distributed to most tissues with the highest concentrations occurring in the kidney, liver and the intestinal contents (preclinical studies). *In vitro* protein binding of tenofovir to plasma or serum protein was less than 0.7 and 7.2%, respectively, over the tenofovir concentration range 0.01 to 25 μg/ml.

Biotransformation

In vitro studies have determined that neither tenofovir disoproxil fumarate nor tenofovir are substrates for the CYP450 enzymes. Moreover, at concentrations substantially higher (approximately 300-fold) than those observed *in vivo*, tenofovir did not inhibit *in vitro* drug metabolism mediated by any of the major human CYP450 isoforms involved in drug biotransformation (CYP3A4, CYP2D6, CYP2C9, CYP2E1, or CYP1A1/2). Tenofovir disoproxil fumarate at a concentration of 100 μmol/l had no effect on any of the CYP450 isoforms, except CYP1A1/2, where a small (6%) but statistically significant reduction in metabolism of CYP1A1/2 substrate was observed. Based on these data, it is unlikely that clinically significant interactions involving tenofovir disoproxil fumarate and medicinal products metabolised by CYP450 would occur.

Elimination

Tenofovir is primarily excreted by the kidney by both filtration and an active tubular transport system with approximately 70–80% of the dose excreted unchanged in urine following intravenous administration. Total clearance has been estimated to be approximately 230 ml/h/kg (approximately 300 ml/min). Renal clearance has been estimated to be approximately 160 ml/h/kg (approximately 210 ml/min), which is in excess of the glomerular filtration rate. This indicates that active tubular secretion is an important part of the elimination of tenofovir. Following oral administration, the terminal half-life of tenofovir is approximately 12 to 18 hours.

Studies have established the pathway of active tubular secretion of tenofovir to be influx into proximal tubule cell by the human organic anion transporters (hOAT) 1 and 3 and efflux into the urine by the multidrug resistant protein 4 (MRP 4).

Linearity/non-linearity

The pharmacokinetics of tenofovir were independent of tenofovir disoproxil fumarate dose over the dose range 75 to 600 mg and were not affected by repeated dosing at any dose level.

Gender

Limited data on the pharmacokinetics of tenofovir in women indicate no major gender effect.

Age

Pharmacokinetic studies have not been performed in children and adolescents (under 18) or in the elderly (over 65).

Ethnicity

Pharmacokinetics have not been specifically studied in different ethnic groups.

Renal impairment

Pharmacokinetic parameters of tenofovir were determined following administration of a single dose of tenofovir disoproxil 245 mg to 40 non-HIV, non-HBV infected patients with varying degrees of renal impairment defined according to baseline creatinine clearance (CrCl) (normal renal function when CrCl > 80 ml/min; mild with CrCl = 50–79 ml/min; moderate with CrCl = 30–49 ml/min and severe with CrCl = 10–29 ml/min).

Compared with patients with normal renal function, the mean (%CV) tenofovir exposure increased from 2,185 (12%) ng·h/ml in subjects with CrCl > 80 ml/min to respectively 3,064 (30%) ng·h/ml, 6,009 (42%) ng·h/ml and 15,985 (45%) ng·h/ml in patients with mild, moderate and severe renal impairment. The dosing recommendations in patients with renal impairment, with increased dosing interval, are expected to result in higher peak plasma concentrations and lower C_{min} levels in patients with renal impairment compared with patients with normal renal function. The clinical implications of this are unknown.

In patients with end-stage renal disease (ESRD) (CrCl < 10 ml/min) requiring haemodialysis, between dialysis tenofovir concentrations substantially increased over 48 hours achieving a mean C_{min} of 1,032 ng/ml and a mean AUC_{0-48h} of 42,857 ng·h/ml.

It is recommended that the dosing interval for tenofovir disoproxil 245 mg (as fumarate) is modified in patients with creatinine clearance < 50 ml/min or in patients who already have ESRD and require dialysis (see section 4.2).

The pharmacokinetics of tenofovir in non-haemodialysis patients with creatinine clearance < 10 ml/min and in patients with ESRD managed by peritoneal or other forms of dialysis have not been studied.

Hepatic impairment

A single 245 mg dose of tenofovir disoproxil was administered to non-HIV, non-HBV infected patients with varying degrees of hepatic impairment defined according to Child–Pugh–Turcotte (CPT) classification. Tenofovir pharmacokinetics were not substantially altered in subjects with hepatic impairment suggesting that no dose adjustment is required in these subjects. The mean (%CV) tenofovir C_{max} and AUC_{0-∞} values were 223 (34.8%) ng/ml and 2,050 (50.8%) ng·h/ml, respectively, in normal subjects compared with 289 (46.0%) ng/ml and 2,310 (43.5%) ng·h/ml in subjects with moderate hepatic impairment, and 305 (24.8%) ng/ml and 2,740 (44.0%) ng·h/ml in subjects with severe hepatic impairment.

Intracellular pharmacokinetics

In non-proliferating human peripheral blood mononuclear cells (PBMCs) the half-life of tenofovir diphosphate was found to be approximately 50 hours, whereas the half-life in phytohaemagglutinin-stimulated PBMCs was found to be approximately 10 hours.

5.3 Preclinical safety data

Preclinical studies conducted in rats, dogs and monkeys revealed target organ effects in gastrointestinal tract, kidney, bone and a decrease in serum phosphate concentration. Bone toxicity was diagnosed as osteomalacia (monkeys) and reduced bone mineral density (rats and dogs).

Findings in the rat and monkey studies indicated that there was a substance-related decrease in intestinal absorption of phosphate with potential secondary reduction in bone mineral density. However, no conclusion could be drawn on the mechanism(s) underlying these toxicities.

Reproductive studies were conducted in rats and rabbits. There were no effects on mating or fertility parameters or on any pregnancy or foetal parameter. There were no gross foetal alterations of soft or skeletal tissues. Tenofovir disoproxil fumarate reduced the viability index and weight of pups in peri-post natal toxicity studies.

Genotoxicity studies have shown that tenofovir disoproxil fumarate was negative in the *in vivo* mouse bone marrow micronucleus assay but was positive for inducing forward mutations in the *in vitro* L5178Y mouse lymphoma cell assay in the presence or absence of S9 metabolic activation.

Tenofovir disoproxil fumarate was positive in the Ames test (strain TA 1535) in two out of three studies, once in the presence of S9 mix (6.2- to 6.8-fold increase) and once without S9 mix. Tenofovir disoproxil fumarate was also weakly positive in an *in vivo* / *in vitro* unscheduled DNA synthesis test in primary rat hepatocytes.

Tenofovir disoproxil fumarate did not show any carcinogenic potential in a long-term oral carcinogenicity study in rats. A long-term oral carcinogenicity study in mice showed a low incidence of duodenal tumours, considered likely related to high local concentrations of tenofovir disoproxil fumarate in the gastrointestinal tract at a dose of 600 mg/kg/day. While the mechanism of tumour formation is uncertain, the findings are unlikely to be of relevance to humans.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Croscarmellose sodium, lactose monohydrate, microcrystalline cellulose, pregelatinized starch, magnesium stearate, hypromellose, titanium dioxide and triacetin.

6.2 Incompatibilities

Not applicable.

6.3 Shelf-life

24 months

6.4 Special precautions for storage

Do not store above 30°C.

6.5 Nature and contents of container

HDPE container containing 30 tablets

6.6 Special precautions for disposal

No special requirements.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. Product Registration Holder:

UNIMED SDN BHD

No. 53, Jalan Tembaga SD 5/2B,
Bandar Sri Damansara, 52200, Kuala Lumpur,
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8. MANUFACTURED BY

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